

# **RC Design**

# **Project 01**

**We can delete this since it no need**



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## 1 How to Design

- Building Load contains : **gravity** and **lateral** Loads.

### 1.1 Convert Area Loads to Line Load on Beams

Slabs distribute their load to beam depending on slab span direction and support condition (Eurocode 2, Section on Slabs [1]; IStructE Manual, Chapter 4 [6]).

#### The Tributary Width Method (IStructE Manual, Chapter 4 [6]):

For a **one-way slab**, load goes to the beams on the short sides:

$$\text{Beam Line Load (kN/m)} = \text{Floor Load (kN/m}^2\text{)} \times \text{Tributary Width (m)}$$

**Tributary Width** = half the slab span on each side of the beam (IStructE Manual, Cl. 3.2 [6]).

**Example:** If Slab spans 5m between two beams, so each beam take  $5/2 = 2.5\text{m}$  tributary width. And if floor load is  $10\text{kN/m}^2$ , then beam line load is  $10 \times 2.5 = 25\text{kN/m}$

For a **two-way slab**, load distributes in both directions (Eurocode 2, Section on Two-Way Slabs [1]; IStructE Manual, Chapter 5 [6]):

- Short span beam take triangular load
- Long span beam take trapezoidal load

### 1.2 Apply ULS and SLS Load Combinations

- ULS (Ultimate Limit State) - for strength design

$$\sum_i \gamma_{G,i} G_{k,i} + \gamma_{Q,1} Q_{k,1} + \sum_{j>1} \gamma_{Q,j} \varphi_{0,j} Q_{k,j}$$

- SLS (serviceability Limit State) - for deflection/crack check

Characteristic Combinations:

$$\sum_i G_{k,i} + Q_{k,1} + \sum_{j>1} \varphi_{0,j} Q_{k,j} + (P_k)$$



Quasi-permanent (for long-term deflection)

$$\sum_i G_{k,i} + \sum_j \varphi_{2,j} Q_{k,j} + (P_k)$$

### 1.3 Beam Support Conditions

Support conditions determine the bending moment diagram shape and therefore where reinforcement must be placed (Eurocode 2, Section on Structural Analysis [1]; IStructE Manual, Chapter 4 [6]):

Table 1.1: Beam support condition

| Support Type            | Description                   | Moment at Support          |
|-------------------------|-------------------------------|----------------------------|
| <b>Simply supported</b> | Beam sits on wall/pad         | Zero moment at ends        |
| <b>Continuous beam</b>  | Multiple spans, monolithic    | Hogging moment at supports |
| <b>Fixed end</b>        | Fully restrained (rare in RC) | Full fixity moment         |
| <b>Partially fixed</b>  | Column/beam junction          | Partial moment transfer    |

| Support Type     | Rotation Allowed | Moment at Support | Typical Structure     |
|------------------|------------------|-------------------|-----------------------|
| Simply Supported | Yes              | 0                 | Beam on columns       |
| Continuous       | Small rotation   | Negative moment   | Multi-span floor beam |
| Fixed            | No rotation      | Maximum moment    | Cantilever            |
| Partially Fixed  | Limited rotation | Partial moment    | Beam-column joint     |

In reinforced concrete buildings, beams are almost always continuous over columns because they are cast monolithically (IStructE Manual, Chapter 4, Cl. 4.2 [6]).

### 1.4 Design Workflow

- **STEP 1:** Establish building layout and use
- **STEP 2:** Identify slab type (one-way / two-way)
- **STEP 3:** Gather  $G_k$ , permanent loads  
(EN 1991-1-1:2002, Annex A, Table A.1 [3])
- **STEP 4:** Gather  $Q_k$ , variable loads  
(EN 1991-1-1:2002, Cl. 6.3.1, Table 6.2 [3])
- **STEP 5:** Apply ULS combination

- **STEP 6:** Convert to beam line load
- **STEP 7:** Calculate bending moments and shear  
(IStructE Manual, Ch. 4 [6])  
(Eurocode 2, Structural Analysis [1])
- **STEP 8:** Size beam using  $L/d$  ratio  
(Eurocode 2, Deflection Section [1])
- **STEP 9:** Design reinforcement  
(Eurocode 2, ULS Design Section [1])
- **STEP 10:** Check SLS — deflection and cracking  
(EN 1990:2002, Cl. 6.5 [5])  
(Eurocode 2, SLS Section [1])
- **STEP 11:** Detail and produce drawings  
(Eurocode 2, Detailing Section [1])  
(IStructE Manual, Ch. 4 [6])

## 1.5 The Key Distinction: what are we Designing

According to (IStructE Manual, Chapter 3 [6]):

| Element    | How Load Works                        |
|------------|---------------------------------------|
| Beam       | Carries only load from its own floor  |
| Column     | Carries load from all floors above it |
| Foundation | Carries load from the entire building |

Table 1.2: How structure work

## 1.6 The Unit Journey

| Unit              | Name        | Used For                 | How We Get It                              |
|-------------------|-------------|--------------------------|--------------------------------------------|
| kN/m <sup>3</sup> | Density     | Material property        | Given in [3] Table A.1                     |
| kN/m <sup>2</sup> | Area load   | Slab design              | Density $\times$ thickness                 |
| kN/m              | Line load   | Beam design              | kN/m <sup>2</sup> $\times$ tributary width |
| kN                | Point force | Column/foundation design | kN/m $\times$ span $\div$ 2                |

Table 1.3: Unit of load to consider in design

## 2 Introduction

hello  $y = ax + b = 2 + 3$  This is Our Team Intro.

### 3 Detailed Design Calculations

#### 3.1 Loading

##### 3.1.1 Live Load

For imposed loads on floor according to Table 6.1 – 6.4 in EN 1991-1-1:2002 [3].  
Is summary in Table 3.1.

Table 3.1: Imposed Loads On Floor Base On EN 1991-1-1:2002

| Category          | Sub-category / Use                                                          | $q_k$<br>[kN/m <sup>2</sup> ] | $Q_k$<br>[kN]       |
|-------------------|-----------------------------------------------------------------------------|-------------------------------|---------------------|
| <b>Category A</b> | Areas for domestic and residential activities                               |                               |                     |
|                   | - Floors                                                                    | 1,5 to 2,0                    | 2,0 to 3,0          |
|                   | - Stairs                                                                    | 2,0 to 4,0                    | 2,0 to 4,0          |
|                   | - Balconies                                                                 | 2,5 to 4,0                    | 2,0 to 3,0          |
| <b>Category B</b> | Office areas                                                                | 2,0 to 3,0                    | 1,5 to 4,5          |
| <b>Category C</b> | Areas where people may congregate                                           |                               |                     |
| C1                | Areas with tables<br>(e.g. schools, cafes, restaurants, reading rooms)      | 2,0 to 3,0                    | 3,0 to 4,0          |
| C2                | Areas with fixed seats<br>(e.g. theatres, cinemas, churches)                | 3,0 to 4,0                    | 2,5 to 7,0<br>(4,0) |
| C3                | Areas without obstacles for moving people<br>(e.g. museums, entrance halls) | 3,0 to 5,0                    | 4,0 to 7,0          |

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Table 3.1: Imposed Loads On Floor Base On EN 1991-1-1:2002

| Category          | Sub-category / Use                                                                  | $q_k$<br>[kN/m <sup>2</sup> ] | $Q_k$<br>[kN]       |
|-------------------|-------------------------------------------------------------------------------------|-------------------------------|---------------------|
| C4                | Areas with possible physical activities<br>(e.g. dance halls, gyms)                 | 3,5 to 5,0                    | 3,5 to 7,0          |
| C5                | Areas susceptible to large crowds<br>(e.g. concert halls, sports halls with stands) | 5,0 to 7,5                    | 3,5 to 4,5          |
| <b>Category D</b> | Shopping areas                                                                      |                               |                     |
| D1                | Areas in general retail shops                                                       | 4,0 to 5,0                    | 3,5 to 7,0<br>(4,0) |
| D2                | Areas in department stores                                                          | 4,0 to 5,0                    | 3,5 to 7,0          |
| <b>Category E</b> | Storage and industrial activities<br>(general storage, accessible areas)            | 7,5                           | 7,0                 |

And this is the imposed loads on second floor that we assumed. which shown in Table 3.2.

Table 3.2: Imposed Loads On Second Floor

| No. | Room / Area                  | Category<br>(EN 1991-1-1) | $q_k$<br>(kN/m <sup>2</sup> ) |
|-----|------------------------------|---------------------------|-------------------------------|
| 1   | Receptionist/<br>Reception   | B                         | 3                             |
| 2   | Workstation /<br>Open office | B                         | 3                             |
| 3   | Meeting Room                 | C1                        | 3                             |
| 4   | President Office             | B                         | 3                             |
| 5   | Vice President Offices       | B                         | 3                             |

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Table 3.2: Imposed Loads On Second Floor

| No. | Room / Area                      | Category<br>(EN 1991-1-1) | $q_k$<br>(kN/m <sup>2</sup> ) |
|-----|----------------------------------|---------------------------|-------------------------------|
| 6   | Accounting                       | B                         | 3                             |
| 7   | General Sale &<br>Administration | B                         | 3                             |
| 8   | Copy /<br>Printer Area           | B                         | 3                             |
| 9   | Storages                         | E1                        | 5                             |
| 10  | Lift Lobby /<br>Lift Area        | C3                        | 4                             |
| 11  | Stair (Main Stair)               | A                         | 4                             |
| 12  | Restroom (Men)                   | B                         | 3                             |
| 13  | Restroom (Women)                 | B                         | 3                             |
| 14  | —                                | —                         | —                             |
| 15  | —                                | —                         | —                             |
| 16  | —                                | —                         | —                             |
| 17  | Entrance                         | C3                        | 4                             |
| 18  | Emergency Stair                  | A                         | 4                             |
| 19  | Server &<br>Technical Room       | E1<br>(conservative)      | 5                             |
| 20  | Kitchen /<br>Pantry              | C1                        | 3                             |
| 21  | Library                          | C3                        | 4                             |
| 22  | Share Space                      | C3                        | 4                             |

### 3.1.2 Dead Load

According to (EN 1990:2002, Cl. 4.1.1 [5]) Permanent loads are defined as loads whose variation in magnitude over time is negligible. The unit weights of construction materials used to calculate these loads are given in (EN 1991-1-1:2002, Annex A, Table A.1 [3]) and (EN 1991-1-1:2025, Annex A [2]):

Table 3.3: Dead Load Typical Value

| Source                                 | Typical Value                        |
|----------------------------------------|--------------------------------------|
| Reinforced concrete slab (200mm thick) | $0.2 \times 25 = 5.0 \text{ kN/m}^2$ |
| Screed/floor finish                    | $1.5 \text{ kN/m}^2$                 |
| Ceiling + services (MEP)               | $0.5 \text{ kN/m}^2$                 |
| Partitions (treated as permanent)      | $1, 0 \text{ kN/m}^2$                |
| <b>Total <math>G_k</math></b>          | $8.0 \text{ kN/m}^2$                 |

## References

- [1] BSI, *BS EN 1992-1-1:2023 - Eurocode 2: Design of concrete structures - Part 1-1: General rules and rules for buildings, bridges and civil engineering structures*, BSI Standards Publication, 2023.
- [2] BSI, *BS EN 1991-1-1:2025 - Eurocode 1: Actions on structures - Part 1-1: Specific weight of materials, self-weight of construction works and imposed loads for buildings*, BSI Standards Publication, 2025.
- [3] CEN, *EN 1991-1-1:2002 - Eurocode 1: Actions on structures - Part 1-1: General actions - Densities, self-weight, imposed loads for buildings*, European Committee for Standardization, 2002. [Online]. Available: <https://www.phd.eng.br/wp-content/uploads/2015/12/en.1991.1.1.2002.pdf>
- [4] CEN, *EN 1991-1-4:2005+A1 - Eurocode 1: Actions on structures - Part 1-4: General actions - Wind actions*, European Standard, Incorporating corrigendum January 2010, April 2010.
- [5] CEN, *EN 1990:2002 - Eurocode: Basis of Structural Design*, European Committee for Standardization, 2002. [Online]. Available: <https://www.phd.eng.br/wp-content/uploads/2015/12/en.1990.2002.pdf>
- [6] Institution of Structural Engineers (IStructE), *Manual for the Design of Concrete Building Structures to Eurocode 2*, 2nd ed., IStructE, London, 2006. [Online]. Available: <https://archive.org/details/manual-for-the-design-of-concrete-building-structures-to-eurocode-2>

*Note that :* We may use the referennces provide by professor (check in telegram group) if it still not enough we can add more maybe.



## Appendices